INCH-POUND
MIL-C-83522/6C
30 December 1988
SUPERSEDING
MIL-C-83522/6B
28 September 1987

MILITARY SPECIFICATION SHEET

CONNECTOR, PLUG, FIBER OPTIC, SINGLE TERMINUS, THREADED, EXPANDED BEAM LENS, EPOXY, FOR 50/125 µm, 62.5 µm, AND 100/140 µm

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-83522.

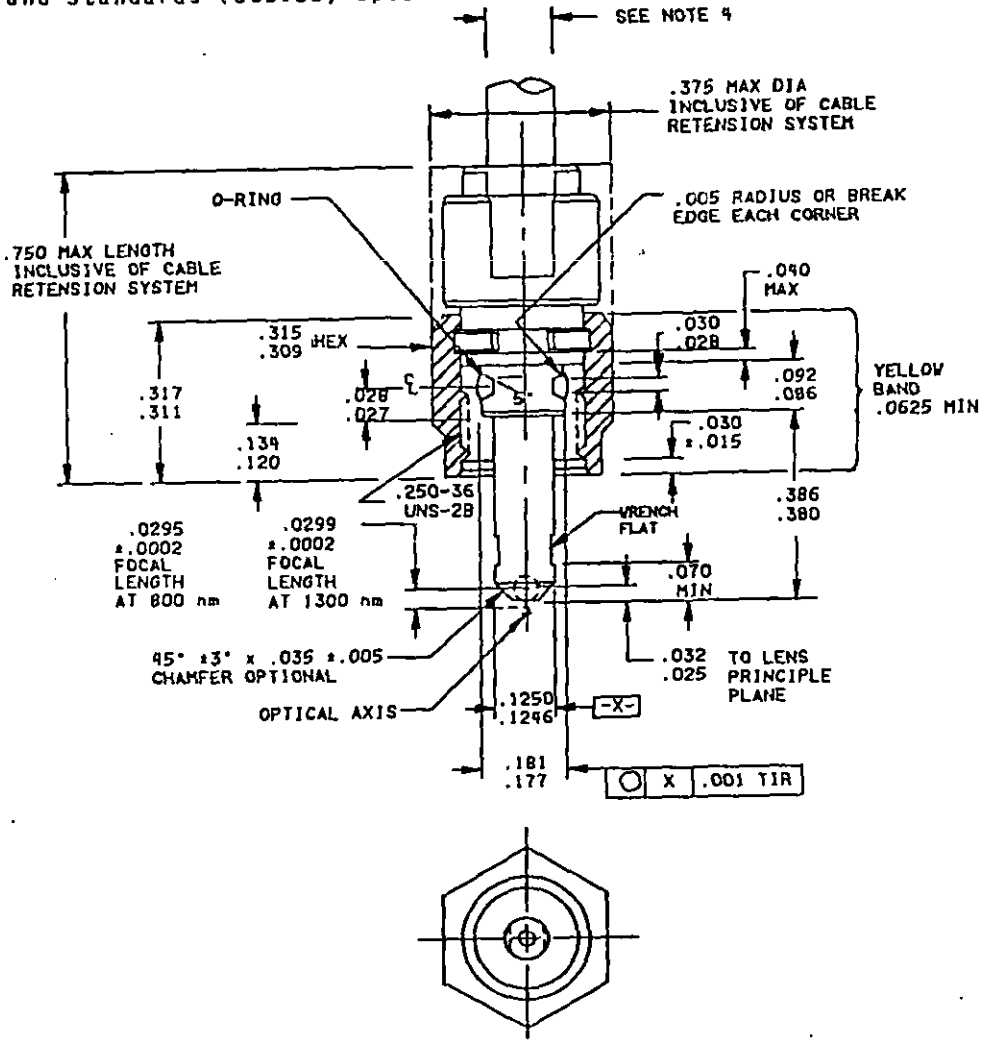


FIGURE 1. Dimensions and configuration.

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Inches	mm	Inches	mm
.0002	0.005	.092	2.34
.001	0.03	.120	3.05
.005	0.13	.1246	3.165
.015	0.38	.1250	3.180
.025	0.64	.134	3.40
.027	0.69	.177	4.50
.028	0.71	.181	4.60
.0295	0.749	.250	6.35
.0299	0.760	.309	7.85
.030	0.76	.311	7.90
.032	0.81	.315	8.00
.035	0.89	.317	8.05
.040	1.02	.375	9.53
.055	1.40	.380	9.65
.0625	1.588	.386	9.80
.070	1.78	.750	19.05
.086	2.18		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Three holes .016 (0.41 mm) minimum diameter, equally spaced, are required for safety wiring after mating. Location of the coupling nuts optional.
4. Dimensions for fiber optic cable shall be in accordance with DOD-C-85045.

FIGURE 1. Dimensions and configuration - Continued.

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REQUIREMENTS:

Metals: The connector housing and other metal components shall be of nickel silver alloy C79200 in accordance with ASTM-B-151.

Epoxies: Use Trabond 230, Eccobond 144B or an equivalent epoxy approved by the qualifying activity.

Dimensions and configuration: See figure 1.

Fiber optic cable requirements: In accordance with D85045/2-1C and M85045/2-10.

Fiber optic termini: In accordance with MIL-F-49291.

Method of optical alignment: Alignment bushing.

Lens configuration: The lens shall be made from an environmentally resistant material, and its principal plane location and focal length shall be as specified on figure 1.

Coating requirements: The lens shall be antireflection coated for operation at the specified wavelength. The residual lens reflection loss per surface after coating shall not exceed 0.1 dB.

Cleaning procedures: If cleaning is necessary, the lens shall be cleanable using lens tissue, a cotton swab, or a camel-hair brush and MIL-C-675 optical antireflective coating cleaning fluid.

Optical requirements:

Number of optical termini: One.

Coupling loss (attenuation): 3 dB maximum per mated pair (steady-state excitation).

Weight: 4 grams maximum.

Polarization: Not applicable.

Safety wire holes: Required.

Force to engage and disengage:

Longitudinal force: 12 pounds maximum.

Torque: 4 inch-pounds maximum.

Coupling proof torque: 5 inch-pounds minimum.

Coupling mechanism retention force: 60 pounds minimum.

Ozone exposure: Not applicable.

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Part or Identifying Number: See table I.

TABLE I. Part or Identifying Number(PIN).

PIN M83522/6-	Fiber size	Operating wavelength
01	50/125	820
02		1350
03	62.5/125	820
04		1350
05	100/140	820
06		1350

Plug accessories: Each plug shall be supplied with suitable means for terminating and clamping a single-fiber optical cable of the fiber size and cable diameter range specified. One protective cable-end cap and one set of assembly instructions shall be included. The assembly instructions shall include a listing of the epoxies specified in this specification sheet and a cleaning procedure statement as follows:

Cleaning procedures: The lens shall be cleaned using lens tissue, a cotton swab, or a camel-hair brush and MIL-C-675 optical antireflective coating cleaning fluid.

NOTE: The plug mates with a MIL-C-83522/4 receptacle; MIL-C-83522/5 receptacle; MIL-C-83522/7 receptacle; MIL-C-83455/8 receptacle; or a MIL-C-83522/3 adapter mated with another MIL-C-83522/6 plug. Connection to any other type plug may result in excessive coupling loss.

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Navy - SH
Air Force - 11, 17, 19, 80, 90, 99
DLA - ES

User activities:
Navy - AS, MC, OS
Air Force - 13, 14

Preparing activity:
Air Force - 85

Agent:
DLA - ES

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